DHARA

A National Legal Knowledge Platform for Bangladesh

Project Proposal

Submitted by

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Table of Contents

Executive Summary **4**
1. Introduction **5**
1.1 Scale of the problem 5
1.2 What the official portal does and does not provide 5
1.3 Two further limitations 5
1.4 Who bears the cost 5
2. Motivation **5**
2.1 Why this is worth solving 5
2.2 Why retrieval-augmented generation 6
2.3 The organising principle 6
2.4 What the project does not claim 6
3. Background Study **6**
3.1 International and regional systems 6
3.2 National resources 6
3.3 The gap this project addresses 6
4. Proposed Solution **7**
4.1 The complete platform 7
4.2 Release sequence and dependencies 7
4.3 Pilot architecture 8
4.4 Provision record 8
4.5 Technology stack 8
5. Scope and Pilot Implementation **9**
5.1 Corpus and versioning 9
5.2 Temporal layer, the core contribution 9
5.3 Retrieval and grounded answering 9
5.4 User-facing features 9
5.5 Out of scope for the pilot 10
5.6 Work plan and timeline 10
5.7 Milestones and monitoring 10
5.8 Risk assessment and mitigation 11
5.9 Applicant capability 11
6. Budget and Resource Plan **12**
6.1 Justification of principal heads 12
6.2 Resources contributed without charge 13
7. Impact on Society **13**
7.1 Social, economic and institutional impact 13
7.2 Alignment with ICT Division priority areas 13
7.3 Alignment with national legal-sector policy 13
7.4 Data sovereignty 14
7.5 Measured, not asserted 14
8. Alignment with Sustainable Development Goals (SDGs) **14**
9. Revenue Plan **14**
9.1 Handover and open licensing 14
9.2 Operating sustainability 14
9.3 Commercial model from v2.0 onward 15
10. Expected Outcomes and Deliverables **15**
10.1 Deliverables 15
10.2 Evaluation methodology 15
10.3 Targets 16
References **16**

Executive Summary

Bangladesh's judiciary carried 46,39,476 pending cases as of 31 March 2026, of which 40,78,432 were in the lower courts. Approximately 2,301 subordinate judges carry that load, about 1,756 cases each. Clearing the subordinate backlog within one year, assuming no new filings, would require roughly 6,300 judges, close to three times the present bench. Recruitment alone cannot close a gap of that scale.

Most causes of backlog are not software problems. One is. Bangladeshi statutory law is difficult to search and impossible to search correctly with respect to time. A dispute is decided by the law as it stood on the date the cause of action arose, yet every system available in the country today, including the official portal, commercial databases and general-purpose chatbots, returns only the current consolidated text.

Dhara is proposed as a national legal knowledge platform: a single environment in which judges, advocates and legal officers research the law, analyse a matter, draft the filing and manage the case, with every legal statement traceable to a dated, verifiable source. The full platform is a multi-year undertaking and is described in Section 4 so that the Division can see where the work leads.

The pilot delivers one layer, completely: the full Bangladesh Code represented at provision level, bilingual, with each provision carrying the dates between which it was in force, served through a cited research interface that abstains rather than guesses. That layer is the prerequisite for everything above it, because drafting from superseded law is worse than not drafting at all, and it is the layer that no commercial party has an incentive to build and release openly.

The project requests BDT 4,85,000 over six months. The budget is almost entirely engineering time, and that is the point rather than an overhead on it. No money is spent re-digitising statutory text that is already public. The money is spent completing that text against the official portal, restructuring it to provision level, and attaching validity dates to every amended provision, which is work that no existing dataset has done. On completion, the structured corpus, the public API and the evaluation benchmark are handed to the ICT Division under Article 21.5 and released under an open licence, and a free Bangla public search interface remains in operation.

1. Introduction

Legal work in Bangladesh is conducted across disconnected tools: a statute portal that shows only present text, paid foreign databases for case law, word processors for drafting, paper diaries for deadlines, and increasingly, general-purpose chatbots that fabricate section numbers. Nothing connects them, and nothing verifies them.

1.1 Scale of the problem

46,39,476	**Cases pending across all levels of the judiciary as of 31 March 2026 [1]**
40,78,432	Of those, pending in the lower courts [1]
About 1,756	Average caseload per subordinate judge [2]
About 6,300	Judges required to clear the subordinate backlog in one year, roughly three times the current bench [2]

Hiring cannot close a threefold gap. Part of the load must be carried by making legal work faster and less error-prone, and that is a software problem.

1.2 What the official portal does and does not provide

The Bangladesh Code portal is comprehensive, free and well maintained, and this proposal treats it as the authoritative public source. It records that a provision was amended. It does not record when.

A typical amendment footnote reads in full:

> *"The words 'and two Taka note' were inserted by section 2 of the Bangladesh Currency (Amendment) Act, 1989."*

The footnote tells you	**The footnote withholds**
That this provision was amended, and which Act amended it	The date that amendment took effect, and what the provision said before it

Both withheld items are required to answer the only question that matters in a historical dispute: what did this provision say on the day the cause of action arose? A claim that arose in 2015 is judged by the 2015 text. The portal displays the 2026 text. Nothing on the page tells you whether those are the same.

1.3 Two further limitations

- Keyword-only retrieval. Bangla is morphologically rich and legal Bangla is highly synonymous. A query phrased around unpaid wages will not reach a provision drafted with an equivalent term, and a Bangla query cannot reach an English-drafted section.
- No programmatic access. Provisions are served as HTML pages. There is no public API, no bulk export and no structured representation, so no downstream system can filter, compare or version them.

1.4 Who bears the cost

A senior advocate in Dhaka absorbs these limitations through experience and paid subscriptions. A junior lawyer in a district town, a legal aid officer and a self-represented litigant cannot. The cost of authoritative legal research tracks income and geography.

A second cost is now emerging. General-purpose language models are already in use for legal questions in Bangladesh. They answer fluently, invent section numbers, and hold no verified knowledge of Bangladeshi amendment history. Measured rates of fabricated citations in such models answering legal queries are high enough to constitute a practical hazard [8].

2. Motivation

2.1 Why this is worth solving

1. Research time is the dominant cost in low-value litigation, which is exactly the class of matter that poor litigants bring.
2. Submissions built on superseded provisions consume hearing time and generate adjournments, a direct contribution to backlog.

3. Most litigants do not know which law or forum governs their dispute, and a claim filed in the wrong forum is dismissed after months.
4. A structured, version-tracked, machine-readable Bangladeshi legal corpus does not exist. Once built, it is reusable public infrastructure for every subsequent legal-technology effort in the country.

2.2 Why retrieval-augmented generation

Keyword search cannot answer a natural-language question or bridge Bangla and English. Direct prompting of a general-purpose model fabricates provisions and offers no provenance. Retrieval-augmented generation is the correct architecture because it separates knowledge, which is retrieved, citable and auditable, from language ability, which belongs to the model.

Plain retrieval-augmented generation is nevertheless insufficient here. It retrieves the current text and is therefore silently wrong for any historical dispute. Adding an explicit temporal layer is this project's contribution.

2.3 The organising principle

Every capability the platform will eventually offer depends on one property: that the system knows, for any provision and any date, what the law actually said. A drafting assistant citing a repealed section produces a defective filing; a precedent analyser relying on an overruled judgment misleads counsel; a limitation calculator built on the wrong commencement date computes the wrong deadline. The substrate therefore comes first, and comes first completely, which is why the pilot covers the entire Bangladesh Code rather than a sample, and why its largest non-personnel head is advocate verification rather than additional features.

2.4 What the project does not claim

Dhara does not give legal advice, does not predict case outcomes and does not profile judges. It is a research assistant with mandatory human review. This is a design decision rather than a disclaimer: predictive judicial analytics is legally restricted in several jurisdictions and is inappropriate for a system of this kind. The project also makes no claim that it will reduce the national case backlog, which is driven principally by judge shortage and procedural delay.

3. Background Study

3.1 International and regional systems

System	Capability	Relevance to Bangladesh
Westlaw (KeyCite), LexisNexis (Shepard's)	Point-in-time statutory text and citation-treatment signalling	Proprietary, priced for Western markets, no Bangladeshi coverage
Harvey AI	Leading generative legal platform, used by major international firms	Built on United States and United Kingdom law. No Bangladeshi statute and no Bangla
Manupatra (Noida, India)	Commercial database including Bangladeshi statutes and case law	Bangladeshi law sold to Bangladeshi firms from abroad, in English, inside an India-first product

The practical position today is that the most usable digital index of Bangladeshi law is maintained outside Bangladesh.

3.2 National resources

Bangladesh-specific legal datasets exist, and this project builds on them rather than duplicating them.

Resource	Provides	Does not provide
BLAD (2026) [5]	1,484 Acts, 35,633 sections, 14,523 footnotes, 1799 to 2025, bilingual, openly licensed	Footnotes stored as unparsed text. Repeal status recorded per Act rather than per section. The authors state that retrieval and temporal analysis remain future work
UKIL-DB-EN (2024) [6]	595 Acts, 18,023 sections, English, with a fine-tuned assistant model	Repealed Acts and sections were deliberately removed during curation, so historical law is absent by construction
bdlaws portal [4]	Official consolidated text, bilingual, with amendment footnotes	No commencement dates. Superseded text not shown. No API or bulk export

3.3 The gap this project addresses

The statutory text of Bangladesh is substantially digitised and in part openly licensed. What does not exist, in any system, public or commercial, domestic or foreign, is a representation in which each provision carries the dates between which it was in force, with retrieval that honours those dates.

That single missing layer separates a legal search tool from a legally correct one, because it is the layer on which the outcome of a historical dispute depends. Building it is the contribution of this pilot.

The existing datasets supply raw statutory text, so no grant money goes into re-digitising what is already public. They do not supply completeness against the official portal, they are not structured to provision level against the schema a temporal system requires, and they carry no dates at all. Closing those three gaps is engineering work, and it is what this grant funds.

4. Proposed Solution

4.1 The complete platform

Dhara is proposed as the connective layer for legal work in Bangladesh: an operating environment in which the law itself is a structured, dated, citable asset rather than a collection of documents to be read. The platform is organised into twelve modules across four releases. The pilot delivers the statutory component of the first release, marked v1.0 below. The remainder is presented for context and is not funded by this grant.

#	Module	Scope	Release
M1	Legal Knowledge Base	Statutes and SROs, then higher-court judgments, subordinate legislation, law reports, regulator circulars	v1.0 to v3.0
M2	Legal Intelligence	Validity intervals and amendment lineage, then citation graph, section-to-judgment index, citation treatment, authority scoring	v1.0 to v3.0
M3	Research and Analysis	Grounded dated question answering and provision browser, then precedent search, issue spotting, research memos	v1.0 to v3.0
M4	Drafting	Pleadings, notices and opinions with live citation checking against the dated corpus	v2.0
M5	Matter Management	Matter records, document repository, chronology, linked research and drafts	v2.0
M6	Calendar and Compliance	Limitation computation, deadline alerts, cause list monitoring, hearing diary	v2.0
M7	Document Intelligence	Bilingual OCR, structured extraction, summarisation, opposing-filing analysis	v2.0 to v3.0
M8	Judicial Tools	Bench memo preparation, bundle summarisation, precedent briefs, consistency checking	v4.0
M9	Institutional Analytics	Backlog composition, ADR eligibility flagging, disposal and pendency trends	v4.0
M10	Public and Legal Aid	Free Bangla search and public API, then grievance triage, forum routing, eligibility screening	v1.0 to v3.0
M11	Platform Services	Identity, roles, chambers, confidentiality controls, audit trail, billing, mobile	v2.0 to v4.0
M12	Trust and Governance	Verification, abstention, provenance, coverage reporting, advocate review, model sovereignty	all releases

4.2 Release sequence and dependencies

Release	Theme	Delivers	Gated by
v1.0 (this pilot)	Statutory foundation	Full Code at provision level, dated; cited research interface; free public search; open corpus and API	Not gated
v1.5	Case law	Higher-court judgments, citation graph, section-to-judgment index, precedent search, citizen grievance triage	Judgment corpus access, an institutional rather than technical dependency
v2.0	Professional workspace	Drafting with live citation checking, matter management, limitation and deadline computation, document intelligence, accounts and billing	v1.0 validity intervals. Without them, drafting cites repealed law

Release	Theme	Delivers	Gated by
v3.0	Analytical depth	Citation treatment, authority scoring, issue spotting, research memos, contract review	v1.5 citation graph
v4.0	Institutional	Judicial tools, court analytics, legal aid intake, mobile and offline	v3.0 treatment classification. A bench memo citing overruled authority is worse than none

Two consequences are worth stating plainly. First, commercial revenue begins at v2.0. Everything before it is infrastructure that no investor would fund and that must nonetheless exist for anything above it to be trustworthy. That is the honest reason to seek public funding for this layer and to hand it back as public infrastructure. Second, the dependency chain is strict rather than conventional: each release is blocked by a correctness property established in the one before it.

4.3 Pilot architecture

Layer	Component	Function
L1	Corpus and versioning	Ingest the open corpus, then resolve amendment footnotes into dated validity intervals with confidence tiering
L2	Indexing	Lexical index and dense multilingual vector index over paired Bangla and English provision text
L3	Temporal hybrid retrieval	Date-filtered candidate generation, sparse and dense fusion, optional cross-encoder reranking
L4	Grounded generation	Citation-attached synthesis, deterministic verification, calibrated abstention

Layers L2 and L4 apply established technique carefully. Layers L1 and L3 contain the contribution. The ordering within L3 is the technical heart of the project: filtering precedes ranking, so a superseded provision is never a candidate and cannot survive into an answer. Reversing those two steps reintroduces precisely the defect the system exists to remove.

4.4 Provision record

Field	Description
provision_id	Stable identifier composed of act_id and section path
text_bn / text_en	Paired bilingual text of the same provision
hierarchy_path	Act, Chapter, Section, Sub-section, Proviso
in_force_from / in_force_to	Validity interval of this version. in_force_to is null where the version is current
amended_by	The amending Act, Ordinance or Statutory Regulatory Order
previous_version_id	Link to the superseded version of the same provision
lineage_confidence	high, medium or none, recording the reliability of the extracted amendment history
footnote_basis	The footnote text from which the commencement date was derived, shown to the user
source_url	Canonical official portal page for the provision

4.5 Technology stack

Category	Selection
Primary corpus	BLAD under CC BY-SA 4.0, with the official Bangladesh Code portal for gap-filling and source links
Supplementary	Bangladesh Gazette: SROs and amending ordinances, for commencement dates
Backend and frontend	Python with FastAPI; Next.js (React) with TypeScript
Database and search	PostgreSQL with pgvector; PostgreSQL full-text search for lexical retrieval
Models	sentence-transformers (multilingual-e5-base, LaBSE fallback); hosted API for generation; open-weight model for self-hosted fallback. No fine-tuning in this phase
Operations	Docker, GitHub, GitHub Actions, single cloud instance

5. Scope and Pilot Implementation

The pilot delivers statutory law only, and delivers all of it. The following is the complete functional specification of what will exist at the end of six months.

5.1 Corpus and versioning

Ref	Requirement
C-1	Ingest the openly licensed structured corpus (1,484 Acts, 35,633 sections) under its licence terms
C-2	Gap analysis against the official portal, with targeted re-scrape only where coverage is missing or malformed
C-3	Cache all raw source pages locally before parsing, so the corpus survives portal change
C-4	Parse to provision level preserving Act, Chapter, Section, Sub-section and Proviso
C-5	Store Bangla and English text of the same provision as a single paired record
C-6	Assign stable, citable provision identifiers of the form act_id/sN@version_date
C-7	Retain the canonical official source URL for every provision
C-8	Produce versioned, reproducible corpus snapshots from a deterministic pipeline

5.2 Temporal layer, the core contribution

Ref	Requirement
T-1	Parse amendment footnotes and extract the amending instrument for each affected provision
T-2	Resolve each amending instrument to its commencement date using the Bangladesh Gazette and Code metadata
T-3	Write in_force_from and in_force_to onto every provision version
T-4	Assign a lineage confidence tier of high, medium or none to each provision
T-5	Link each version to its predecessor, so the superseded wording is retrievable
T-6	Store the footnote text from which the date was derived and display it to the user as provenance
T-7	Mark a provision as date unknown rather than assign a date that cannot be verified
T-8	Generate a word-level difference between consecutive versions of a provision
T-9	Report amendment coverage by confidence tier as a measured figure, validated against a manual audit

5.3 Retrieval and grounded answering

Ref	Requirement
R-1	Accept a natural-language query in Bangla or English together with an as-of date, defaulting to today
R-2	Restrict candidates by validity interval at the database level before any ranking occurs
R-3	Run lexical (BM25) and dense multilingual retrieval in parallel over the filtered set
R-4	Merge the two ranked lists by reciprocal rank fusion, requiring no score calibration
R-5	Support cross-lingual matching through a shared embedding space, without machine translation
G-1	Constrain generation to retrieved provisions and their metadata only
G-2	Attach a provision-level citation to every generated sentence
G-3	Identify the exact span within the source provision that supports each cited sentence
G-4	Verify deterministically that every cited identifier appears in the retrieved set, and withhold any response that fails
G-5	Abstain with an explicit insufficient-grounding response below a calibrated confidence threshold
G-6	Refuse out-of-scope requests such as legal advice, outcome prediction and judicial profiling, and redirect to a legal question
G-7	Warn when the cited provision was amended after the selected date, or when lineage is unavailable

5.4 User-facing features

Feature	Description
Dated research interface	Query in Bangla or English with an as-of date. Answer language is selectable independently of query language
Visible retrieval pipeline	The user watches date filtering, retrieval and citation verification execute, rather than being asked to trust them
Inline citation chips	Each cited sentence carries a numbered chip. Selecting it highlights the exact supporting span in the source panel
Source panel	Every provision the answer draws on, with the version used, and a statement that nothing beyond the list was consulted
Version timeline	A visual track of a provision's versions with the selected date marked, making the effect of the date immediately legible
Provision browser	Navigate by Act and chapter. View version history, full text, and a word-level difference between versions
Provenance disclosure	For every provision: identifier, validity dates, amending instrument, previous version, confidence tier, the footnote the date came from, and the official source link
Four response states	Answered, historical version with warning, coverage warning, and explicit refusal, each visually distinct
Query history	Previous questions with the date used and the outcome, re-runnable
Free Bangla public access	The same engine, with no authentication, permanently free
Public read API	Documented programmatic access over the versioned corpus for legal aid bodies and developers

5.5 Out of scope for the pilot

Judgment corpus and precedent search; citation-treatment classification; drafting; matter management; limitation computation; document upload and OCR; judicial tools; institutional analytics; authentication and billing; mobile applications; any model fine-tuning. These belong to later releases and are not funded by this grant.

5.6 Work plan and timeline

The project runs for six months from first disbursement across three parallel tracks. Six months rather than four reflects the real cost of the amendment-lineage work: resolving commencement dates across more than a thousand Acts is iterative, and the result must be complete and stable before evaluation and advocate review can begin. The final phase is feature-frozen and reserved for release, field testing and handover.

Phase	Data and ML	Backend	Frontend
1 (Months 1-2)	Corpus ingestion, gap analysis, targeted re-scrape, provision schema	Database schema, API skeleton, ingestion endpoints	Wireframes, application scaffold, component library
2 (Months 2-3)	Amendment-footnote parser, Gazette date resolution, confidence tiering, embeddings	Lexical and dense index, corpus query layer	As-of-date query interface, provision browser
3 (Months 3-4)	Retrieval tuning, fusion weighting, reranker assessment	Temporal retrieval endpoint, grounded generation, citation verification, abstention calibration	Answer view, citation chips, coverage warnings
4 (Month 5)	Evaluation set authoring, 200-provision lineage audit, evaluation and baseline comparison	Corrections from advocate review, caching, performance tuning	Amendment difference view, query history
5 (Month 6)	Results tables, benchmark release, final report	Integration, public API documentation, no new features	Bangla public interface, accessibility pass, release

5.7 Milestones and monitoring

ID	Completion criterion	Timing	Evidence submitted
M1	Corpus ingested, schema populated, raw sources cached	End of Month 2	Record counts, schema documentation

ID	Completion criterion	Timing	Evidence submitted
M2	Validity intervals written, coverage measurable	End of Month 3	Coverage by tier, expenditure statement (first tranche report)
M3	End-to-end cited answers with verification and abstention	End of Month 4	Demonstration build, verification logs
M4	Evaluation, lineage audit and advocate review complete	End of Month 5	Metrics, baseline comparison, advocate sign-off
M5	Public release and handover to the ICT Division	End of Month 6	Handover package, public URLs, final expenditure statement

M1 and M2 both fall within the first three months and constitute the verifiable output of the first tranche. All expenditure is supported by receipts and reconciled against the head-wise budget. Deviations between heads exceeding ten percent are reported in advance. The applicants accept the assignment of a mentor under the programme and will participate in evaluation as required.

5.8 Risk assessment and mitigation

Risk	Severity	Mitigation
A user relies on an incorrect answer and suffers legal detriment	High	No legal advice is given and this is stated prominently. Outputs are advocate-verified before release. The system abstains below threshold. Every answer links to the official source. An audit log records every answer served
Commencement dates unrecoverable for a substantial share of amendments	High	Confidence tiering: provisions whose dates cannot be established are marked as date unknown and fall back to current text with a visible warning. Coverage is reported as a measured result. The project succeeds with partial coverage honestly declared
The printed Bangladesh Code, not the portal, is the legally authoritative text	Medium	The system is positioned throughout as a research aid rather than legal authority. Every citation links back to the official portal. This limitation appears in the interface, not only in documentation
Practitioners do not adopt or change established workflow	Medium	The advocate panel becomes the first expert user group. Distribution runs through legal aid offices and law faculties rather than commercial outreach. The free public layer creates usage independent of professional adoption
Source portal changes structure or blocks automated access	Medium	The project builds primarily on an already-downloaded open dataset. All pages needed for gap-filling are cached in Phase 1, and the parser is decoupled from the fetcher
Retrieval quality degrades at full-Code scale	Medium	The hybrid design means lexical search continues to carry exact-term and section-number queries where dense retrieval underperforms on legal Bangla. Both retrievers are evaluated separately so degradation is attributable
Cost overrun with no contingency head in the budget	Low	Compute and API heads are already conservative, the embedding job is chunked and resumable, and a smaller embedding model is available as fallback. Any residual overrun is absorbed by the applicants rather than claimed
Team continuity, as both applicants complete their degree during the period	Medium	All deliverables are handed over and openly licensed, so the corpus and benchmark survive independently of the team. Documentation and deployment instructions form part of the handover

5.9 Applicant capability

Applicant	Background and responsibility
Sazidur Rahman Mahin (Roll 1414)	Final-year Software Engineering student, Institute of Information Technology, University of Dhaka. Software engineering experience at Brain Station 23. Responsible for corpus pipeline, amendment lineage extraction and evaluation

Applicant	Background and responsibility
Shah Alam Abir (Roll 1439)	Final-year Software Engineering student, Institute of Information Technology, University of Dhaka. Software engineering experience at Cefalo Bangladesh Ltd. Published research including a vision-language navigation benchmark. Responsible for retrieval architecture, grounded generation and system integration

- Two national AI hackathon wins between the applicants.
- Prior construction and deployment of retrieval-augmented systems of the same architectural class as the one proposed, including hybrid retrieve-then-rerank pipelines in production.
- Affiliation with active research groups, with published work evaluated against benchmarks.
- A working interactive prototype of the research interface and provision browser has already been built and is available for demonstration.
- Academic supervision and institutional support from the Institute of Information Technology, University of Dhaka.

6. Budget and Resource Plan

The total requested is BDT 4,85,000 (four lakh eighty-five thousand taka only), disbursed over six months within the Small tier of the Innovation Fund.

This budget is almost entirely engineering time, and that is deliberate. The deliverable of this project is a dataset, and a dataset is built by people rather than by hardware. No money is spent re-digitising statutory text that is already freely available. The money is spent completing that text against the official portal, restructuring it to provision level, resolving commencement dates onto every amended provision, building the retrieval layer above it, and verifying the result with practising advocates.

Head of expenditure	Basis of estimate	Amount (BDT)	%
Development stipend	2 engineers, 64 hours per month each, 6 months, at BDT 500 per hour (768 hours in total)	3,84,000	79.2
Advocate verification panel	2 practising advocates, 20 hours each, at BDT 1,200 per hour (40 hours in total)	48,000	9.9
Field testing with legal aid offices	2 districts: travel, facilitated sessions, materials	18,000	3.7
LLM API credits	Generation during development, repeated evaluation runs, and pilot operation	15,000	3.1
Cloud hosting, vector database, public API	Single instance, about eight months	10,000	2.1
GPU compute	Corpus embedding, reranking, self-hosted open-weight fallback	10,000	2.1
Total		**4,85,000**	**100.0**

6.1 Justification of principal heads

Development stipend, BDT 3,84,000. Two engineers at 64 hours per month each over six months, at BDT 500 per hour. The grant therefore buys 768 measured engineering hours at a rate below commercial cost for engineers of comparable experience in Dhaka. Sixty-four hours a month is about fifteen hours a week, which is sustainable alongside the applicants' existing employment and is what they can honestly commit.

Personnel accounts for 79 percent of the request, and that share is the project rather than an overhead on it. The corpus is only partially built. An openly licensed dataset supplies raw statutory text, but it is incomplete against the official portal, it is not structured to provision level against the schema a temporal system requires, and it carries no dates. Closing those gaps is the work: ingestion and gap analysis, the provision parser, the amendment-footnote parser, Gazette commencement-date resolution, confidence tiering, and the retrieval and verification layers built above them. Those are engineering hours, and they produce the asset that is handed back to the Division.

Advocate verification panel, BDT 48,000. Two practising advocates at 20 hours each, at prevailing consultation rates. Forty hours of professional time supports review of a stratified sample of system outputs across legal domains, together with sign-off on the evaluation benchmark. It does not support exhaustive review of the full corpus, and this proposal does not claim that it does. Legal software that no qualified lawyer has examined should not be released to the public, and a benchmark not reviewed by practitioners carries limited weight. The panel also constitutes the project's first expert user group.

LLM API credits, BDT 15,000. A cited answer grounded in about five retrieved provisions costs on the order of BDT 3 at current commercial rates. The allocation covers development iteration, repeated evaluation runs across the 150-question set, advocate review cycles and operation during the field pilot. Retrieved provision text is cached between queries, so repeated questions touching the same sections cost a fraction of the first call.

GPU compute, BDT 10,000. Three uses: one-off embedding of about 35,000 sections, reranking during development and evaluation, and standing up the self-hosted open-weight inference path. The third is the substantive justification, since a project intended to keep national legal infrastructure in Bangladeshi hands cannot route its free public layer through a foreign commercial API with no alternative.

Hosting, BDT 10,000. The complete vector index is of the order of 110 MB, so the system runs on a single modest instance rather than on managed infrastructure. The allocation covers the six-month project period and approximately two months beyond launch, after which continuation depends on the handover arrangement with the Division or on an institutional agreement.

6.2 Resources contributed without charge

Development machines, engineering hours beyond the 768 funded, the raw statutory text under open licence, and academic supervision and institutional affiliation from the Institute of Information Technology, University of Dhaka. There is no contingency head; any overrun is absorbed by the applicants.

7. Impact on Society

7.1 Social, economic and institutional impact

- Social. Litigants and practitioners outside Dhaka gain correctly dated statutory information without a paid subscription. Legal aid organisations can triage more intake with the same staff. Law students and junior advocates gain a tool that teaches correct citation practice, since every answer is anchored to a provision and a date.
- Economic. Lower research cost per matter makes small claims economically viable to pursue. Fewer submissions built on superseded provisions means fewer wasted hearings and adjournments.
- Institutional. The version-tracked corpus is a reusable public asset. Released openly, it lowers the entry cost for every future Bangladeshi legal-technology effort, much as open map data lowers the cost of building navigation software.

7.2 Alignment with ICT Division priority areas

Priority area	Contribution
Public service innovation	Free Bangla public access to correctly dated statutory information, a documented public API usable by legal aid bodies, and the corpus and API handed to the Division as reusable national infrastructure
Socio-economic development	Reduces the cost of authoritative legal research, which currently tracks income and geography, and serves district practitioners and self-represented litigants who have no subscription alternative
ICT industry growth	Establishes Bangla legal NLP capability and an open benchmark in a domain with no existing national resources, lowering the entry cost for every subsequent Bangladeshi legal-technology effort

7.3 Alignment with national legal-sector policy

The Government has committed to expanding mandatory pre-litigation mediation. Under the amended legal aid framework, mediation is now mandatory in most scheduled-law cases, with certified settlements carrying the status of court decrees. The programme currently operates in twenty districts, where it has reduced regular case filings by 62.83 percent, and nationwide expansion is planned [3].

Every mediation requires both parties to understand what the law provides. Where that understanding is absent, settlements are unbalanced or do not occur. Dhara's free public layer addresses that requirement directly. The project therefore supports a policy direction the Government has already adopted rather than proposing a new one.

7.4 Data sovereignty

At present the most usable digital index of Bangladeshi statutory and case law is maintained by a commercial entity outside Bangladesh, sold in foreign currency, within a product designed for another jurisdiction and operating only in English. A nation's legal corpus is public infrastructure of the same character as its mapping or identity data. This project establishes a Bangladeshi-owned, openly licensed alternative and places it in the custody of the ICT Division.

7.5 Measured, not asserted

The project commits to three measurements rather than three claims: a timed user study with law students against manual portal search, the measured accuracy gap over the temporally naive baseline on date-sensitive questions, and a hallucinated-citation rate at or near zero established through deterministic verification. It makes no claim that it will reduce the national case backlog. The honest claim is narrower: the system removes one avoidable source of delay and one avoidable source of error.

8. Alignment with Sustainable Development Goals (SDGs)

Dhara supports five of the seventeen Goals. Alignment with SDG 16 is direct and primary.

SDG	Goal	Contribution
16	Peace, Justice and Strong Institutions	Primary alignment. Makes the statutory corpus searchable, correctly dated and citable, directly expanding access to justice, and produces machine-readable legal infrastructure that courts and legal aid bodies can build on
10	Reduced Inequalities	Narrows the legal capability gap that currently tracks income and geography between senior urban counsel and district practitioners, legal aid officers and self-represented litigants
9	Industry, Innovation and Infrastructure	Builds national digital public infrastructure and advances Bangla-language natural language processing in a specialised domain with no existing resources
4	Quality Education	The open corpus and evaluation benchmark are directly usable in legal education and in NLP research training at Bangladeshi universities
8	Decent Work and Economic Growth	Full-Code coverage includes labour and industrial legislation, helping workers and employers establish entitlements under the law as it stood at the relevant time

9. Revenue Plan

9.1 Handover and open licensing

In accordance with Article 21.5, the structured corpus, public API, evaluation benchmark and documentation are handed to the ICT Division or its designated institution for public service use on completion, and additionally released under an open licence so they remain usable by Bangladeshi researchers, legal aid organisations and developers. The project is designed around this obligation rather than in tension with it: handover means government custody and permanent distribution, which is the best available outcome for infrastructure of this kind.

9.2 Operating sustainability

- Low marginal cost. The retrieval layer runs on commodity hardware and the index is of the order of 110 MB. Only generation depends on a paid service, and the self-hosted path funded here makes that dependency replaceable.
- Hosting beyond the project. The index is small enough to run on a single modest instance, and hosting is funded to roughly two months past launch. Continuation beyond that depends on the handover arrangement with the Division or on an institutional service agreement, both of which are low-cost at this scale.

- Academic continuity. Corpus expansion, judgment ingestion and benchmark maintenance map onto undergraduate and postgraduate theses at the applicants' institution, sustaining development beyond any single funding cycle.
- Institutional anchor. A single service agreement with a legal aid body or judicial training institution would cover baseline operating cost while supplying the domain feedback loop the system needs.

9.3 Commercial model from v2.0 onward

A professional product layer is intended above the public corpus from v2.0. This is declared here so that the boundary is unambiguous: nothing in the table below is funded by this grant, and nothing in it forms part of the handover. Public statutory search remains free in every scenario.

Stream	Customer	Indicative pricing
Public statutory search	General public, legal aid bodies	Free, permanently
Individual subscription	Practising advocates	BDT 1,200 per seat per month
Chamber subscription	Law firms and chambers	BDT 4,000 per month, 4 seats
Enterprise licence	Banks, corporate legal departments, insurers	BDT 25,000 and above per month
Institutional licence	Law and Justice Division, NLASO, judicial academy, Bar Council, law faculties	Annual contract
Regulatory-change monitoring API	Banks, non-bank financial institutions, telecom operators	Per-call or annual contract

The regulatory-monitoring stream deserves particular note. The amendment-tracking engine built for statutes applies unchanged to Bangladesh Bank circulars, securities directives and telecommunications regulations, which are frequently changing instruments that regulated institutions are obliged to track. It reuses the core technology with minimal additional engineering and addresses a segment with existing compliance budgets, making it the most plausible early commercial line.

10. Expected Outcomes and Deliverables

10.1 Deliverables

#	Deliverable	Description
1	Structured national corpus	Full Bangladesh Code at provision level, bilingual, with validity intervals and confidence-tiered lineage
2	Temporal retrieval engine	Hybrid lexical and dense retrieval with date filtering applied before ranking
3	Grounded answering service	Citation-attached generation with deterministic verification and calibrated abstention
4	Web application	Research interface and provision browser as specified in Section 5.4
5	Free Bangla public interface	Open public access to dated statutory information, permanently free
6	Public read API	Documented, rate-limited and openly accessible
7	Evaluation benchmark	150 annotated questions across at least eight legal domains, about 30 date-sensitive, released openly
8	Results report and repository	Metrics, baseline comparison, error analysis, lineage audit and a reproducible codebase
9	Handover package	Corpus, API, benchmark, documentation and deployment instructions delivered to the ICT Division

10.2 Evaluation methodology

The system is evaluated against a 150-question set, each item annotated with its gold provision, spanning at least eight legal domains so that performance is not an artefact of a single area of law. About thirty questions are date-sensitive: they resolve to different provisions depending on the as-of date and are answerable correctly only by a temporally aware system. Twenty are unanswerable, so that abstention is scored as correct only where no governing provision exists and cannot be gamed.

A temporally naive baseline, being the identical pipeline with date filtering disabled, is run on the same set. The accuracy gap on the date-sensitive subset is the experimental result the pilot is designed to produce. Amendment lineage is separately audited against a manually verified sample of 200 provisions, with coverage and precision reported together.

10.3 Targets

Indicator	Target	Verification
Temporal correctness on date-sensitive subset	At least 0.85	Evaluation set against gold provisions
Retrieval Recall@5 over the full Code	At least 0.75	Evaluation set
Citation precision	At least 0.85	Advocate review of sampled answers
Hallucinated-citation rate	At most 0.05	Deterministic verification log
Amendment coverage (high-confidence lineage)	At least 0.60	Automated count
Lineage precision on audited sample	At least 0.90	Manual audit of 200 provisions
Median end-to-end response time	At most 5 seconds	Instrumented logs

These are targets set in advance of implementation. They are not results already obtained. The result of primary interest is the gap between Dhara and the temporally naive baseline on date-sensitive questions. A large gap demonstrates that amendment awareness is not a cosmetic feature but a correctness requirement for legal retrieval, which is the argument this project exists to make.

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